

Prediction and machine learning

[EC524/424]

Winter 2026 Syllabus

<https://github.com/edrubin/EC524W26>

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	Start subject with "EC524 :".	
	PLC 530	PLC 525
	Tu. 3:30p-4:30p	TBA
	edrub.in	jose-rojas-fallas.quarto.pub

Email note: We will do our best to respond promptly to your emails. Our responses may be slower over weekends/holidays. There may be times that our responses take up to 48 hours. Please do not repeatedly send the same email unless it has been more than 48 hours.

	<u>Lecture</u>	<u>Lab</u>
	Tu. & Th., 10:00a-11:20a	Fr., 10:00a-10:50a
	105 Esslinger	072 PLC
	Ed	Jose Ed
	Our class: https://github.com/edrubin/EC524W26/	
	2025: https://github.com/edrubin/EC524W25/	
	2024: https://github.com/edrubin/EC524S24/	
	2023: https://github.com/edrubin/EC524W23/	
	2022: https://github.com/edrubin/EC524W22/	
	2021: https://github.com/edrubin/EC524W21/	
	2020: https://github.com/edrubin/EC524W20/	

Course summary

Description Following the first course on econometrics and causal inference in our sequence, EC524 turns to examining the **tools available and best practices for predicting outcomes**. Put simply, we are now focusing on $\hat{y}_i$ rather than $\hat{\beta}$ from the model $y_i = \alpha + \beta x_i + \varepsilon_i$.

Learning statistical programming is inherent to practicing applied econometrics. Consequently, throughout this course we will also teach the statistical programming language R.

Objectives

1. **Distinguish** between settings that require **causal inference** vs. settings that want **prediction**.
2. Understand the main **themes and best practices** in modern **prediction** methods/contexts.
3. Develop **familiarity** with common machine-learning algorithms—and their strengths/weaknesses.
4. Build **intuition** for prediction—especially the bias-variance tradeoff.
5. Expand **R expertise**.

Prerequisites This course requires the previous course in our sequence—*i.e.*, Economics 423/523. I also assume you are comfortable in R.

Books

I know you are busy and reading for class is often difficult. However, **if you are actually here to learn, then read these books**.

Note Each book (except two of the recommended books) is available for **free online**. The physical copies are also very reasonably priced—I suggest you buy physical versions for books that you like.

Required books

1. **Introduction to Statistical Learning** *ISL*
2. **The Hundred-Page Machine Learning Book** *100ML*
3. **Data Visualization** *Data Viz*

Suggested books

1. **R for Data Science** *RDS*
2. **Introduction to Data Science** *IDS* (not available without purchase)
3. **The Elements of Statistical Learning** *ESL* (the big brother of *ISL*)
4. **Data Science for Public Policy** *DSPP* (e-book available via UO library)

Software and tools

- We will use the statistical programming language **R**.
- We will use **RStudio** to interact with R.

Learning R will require time and effort, but it is a powerful and versatile tool that is valued by many employers. Put in the requisite effort and time, and you will be rewarded.

Labs, assignments, projects, and exams

Attend the lab This course includes a lab, which is **integral to learning** the material in (and passing) this course. The lab includes both general econometrics instruction and computing resources necessary to complete the course and learn/-master its topics.

Assignments

- You will submit **typed assignments via Canvas**, generally in one of two formats (we'll tell you what we want):
 1. An **R notebook** that is hosted somewhere on the web or submitted as a self-contained HTML file;
 2. A link to a **Kaggle notebook**.
- We will grade on a **complete/incomplete scale**.
Low-quality work will be returned to be re-submitted as late.

Late submissions Students whose assignments are occasionally late will be penalized half a letter grade. Students whose assignments are frequently late will be penalized a full letter grade.

Group work Feel free to work together on the assignments. Unless explicitly stated, each student is required to write and submit independent answer sheets. This means that word-for-word copies will not be accepted and will be viewed as academic dishonesty. If you work with other students, you must list the students in your study group at the top of your assignment. If you fail to do so, you will receive a score of zero.

Project 1: Application (Due 04 March 2026) For this project, you will apply what you've learned in this course to a prediction problem/context of your choice. You will:

- Find data for your problem,
- Clean data as needed,
- Use multiple prediction models and best practices to make predictions,
- Write up everything in a clean notebook/blog,
- Present a five-minute presentation on the process/results.

Project 2: Extension (Due 11 March 2026) This project pushes you to extend your knowledge to a new method. You will choose a topic that we have not covered in class—but that is related to topics covered in class—e.g., spectral clustering or time-series prediction. You will:

- Learn about this topic on your own,
- Write a ‘wiki’ that explains this topic using *math and examples*,
- Make and give a five-minute presentation on the topic.

No duplicates for topics. I will provide a list of ideas on the course site. The idea here is that you extend/apply the course’s ideas to situations that **interest you**.

Exams We will proctor an **in-person final** on Monday, March 17th, 2026 from 8:00a–10:00a Pacific. We **will not** offer remote or make-up options for the exam.

Recommendations

1. **Be kind.**
2. **Take responsibility** for your own education and try to **learn** as much as you can.
3. **Do your own work.**
4. Develop your **intuition**—e.g., why would method x work in one situation and fail in another?
5. **Learn R.** Struggle while you try—and use **Google** to figure things out.
6. Come to **office hours**.¹

Honesty and academic integrity

You must do your own work. Do not claim credit for any work other than your own. **Your work should not be identical to others’ work.** Cheating or plagiarizing of any sort on any component of this class will result in a failing grade for the term and a report of the offense to the university. Please acquaint yourself with the [Student Conduct Code](#).

Large language models (and other sources): ChatGPT, GitHub Copilot, Claude, and other related AI ‘assistants’ are great tools. I am totally fine with you using them—and even encourage it. However, you still need to submit work **in your own words**, and you need to **understand the code** that you submit. Anything less is plagiarism, lazy, and a loss of opportunity to actually learn valuable material/tools.

Accessibility

The University of Oregon and I are dedicated to fostering inclusive, equitable, and accessible learning environments for all students. The Accessible Education Center (AEC) assists students with disabilities in reducing barriers in the educational experience. You may be eligible for accommodations for a variety of disabilities – apparent disabilities, such as a mobility or physical disability, or non apparent disabilities, such as chronic illnesses or psychological disabilities. **If you have or think you have a disability** and experience academic barriers, please contact the Accessible Education Center (Location: 360 Oregon Hall; 541-346-1155; uoaec@uoregon.edu) to discuss appropriate accommodations or support. The details of your disability will be kept confidential with the AEC and you are not expected to share this information with others. However, I invite you to discuss any approved accommodations or access needs at any time with me.

¹Two related articles from NPR on office hours: [College Students: How to Make Office Hours Less Scary](#) and [Uncovering A Huge Mystery Of College: Office Hours](#).

Grading

Grades will be assigned as follows.²

<u>Grade</u>	<u>Assignments</u>	<u>min(Projects)</u>	<u>Final exam</u>
A	<i>Incomplete</i> on 0 assignments.	$\geq$ <i>Professional</i>	$\geq$ 80%
B	<i>Incomplete</i> on $\leq$ 1 assignments.	$\geq$ <i>Minor revision</i>	$\geq$ 70%
C	<i>Incomplete</i> on $\leq$ 2 assignments.	$\geq$ <i>Moderate revision</i>	$\geq$ 60%
D	<i>Incomplete</i> on $\leq$ 3 assignments.	$\geq$ <i>Major revision</i>	$\geq$ 50%

Recall that assignments are graded as *Complete* vs. *Incomplete*—the standard for *Complete* is much higher than simply submitting.

²Undergraduates are allowed to miss one additional assignment in the scheme.

Tentative, overly-ambitious, predicted outline

Note: Stay up to date on our class [class's Github page](#).

0. An introduction to prediction and statistical learning

1. What are we doing? **Readings** *ISL* Introduction, Ch1
2. Prediction vs. causal inference **Readings** *Prediction Policy Problems* by Kleinberg *et al.* (2015)
3. Modeling decisions and assessment **Readings** *ISL* Ch3

1. Exploratory data analysis

1. Building insights from graphics **Readings** *Data Viz* Preface, Ch1
2. `ggplot2` **Readings** *Data Viz* Ch3

2. Supervised learning

1. An introduction to machine learning **Readings** *100ML* Preface, Ch1–Ch4; *ISL* 2.1–2.2
2. Resampling methods and other best practices **Readings** *100ML* Ch5; *ISL* Ch5
3. Why don't we stick with regression? **Readings** *ISL* Ch3
4. LASSO and Ridge regression **Readings** *ISL* 6.1–6.3, 6.6
5. Classification and logistic regression **Readings** *ISL* 4.1–4.3
6. Decision trees **Readings** *100ML* 3.3; *ISL* 8.1
7. Ensembles: Bagging, random forests, boosting **Readings** *ISL* 8.2–8.3 *100ML* 7.5 and Ch8
8. SVM **Readings** *100ML* 3.4; *ISL* 9.1–9.4
9. Neural nets **Readings** *100ML* 6
10. Additional topics **Readings** *100ML* Ch7 and Ch11

3. Unsupervised learning

1. Introduction to unsupervised learning **Readings** *100ML* Ch9; *ISL* 10.1
2. Principal components analysis **Readings** *ISL* 10.2; *100ML* 9.3
3. Nearest-neighbor matching, *K*-means, and hierarchical clustering **Readings** *100ML* Ch9; *ISL* 10.3

4. Extensions

1. Bias and fairness **Readings** [Hao \(2019\)](#)